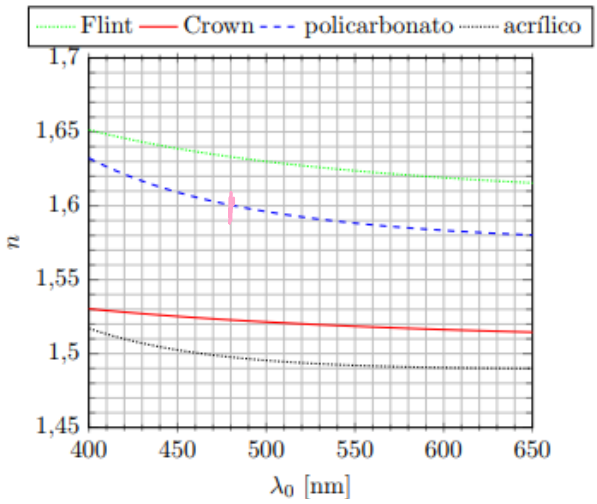
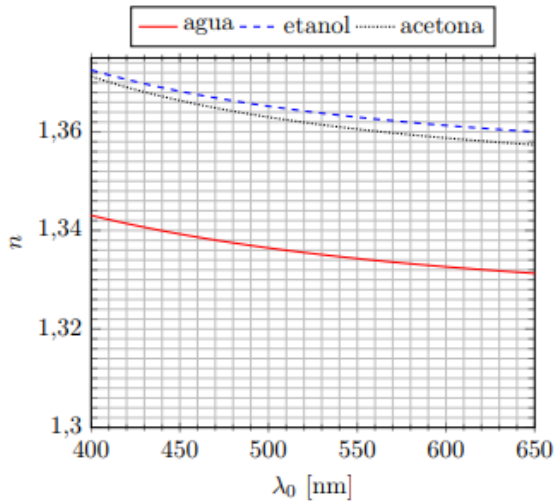




$$n(\lambda_0) = A + \frac{B}{\lambda_0^2} + \frac{C}{\lambda_0^4} + \dots$$

Material	A	B [nm^2]	C [nm^4]
Flint	1,59590	$7{,}8829\,10^3$	$1{,}59477\,10^8$
Crown	1,50380	$4{,}6798\,10^3$	$-7{,}00915\,10^7$
policarbonato	1,57281	$-7{,}80383\,10^2$	$1{,}64800\,10^9$
diamante	2,380260	$1{,}21822\,10^4$	$1{,}8860\,10^8$
cuarzo	1,44804	$3{,}64576\,10^4$	$-1{,}88116\,10^7$
acrílico	1,49743	$-6{,}99016\,10^3$	$1{,}62200\,10^9$
agua	1,32333	$3{,}49329\,10^3$	$-5{,}39345\,10^7$
etanol	1,35265	$3{,}06000\,10^3$	$2{,}00000\,10^7$
acetona	1,34979	$3{,}06000\,10^3$	$6{,}00000\,10^7$

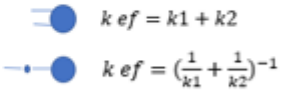
$\operatorname{sen}(\alpha \pm \frac{\pi}{2}) = \pm\cos\alpha$
$\cos\alpha\ \cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$
$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$
$\operatorname{sen}\alpha + \operatorname{sen}\beta = 2\operatorname{sen}\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$
$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \operatorname{sen}\alpha\operatorname{sen}\beta$

$\cos(\alpha \pm \frac{\pi}{2}) = \mp\operatorname{sen}\alpha$
$\cos\alpha\cos(\alpha+\beta) = \frac{1}{2}[\cos(2\alpha+\beta) + \cos\beta]$
$\cos\alpha - \cos\beta = -2\operatorname{sen}\left(\frac{\alpha+\beta}{2}\right)\operatorname{sen}\left(\frac{\alpha-\beta}{2}\right)$
$\operatorname{sen}\alpha - \operatorname{sen}\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\operatorname{sen}\left(\frac{\alpha-\beta}{2}\right)$
$\operatorname{sen}(\alpha \pm \beta) = \operatorname{sen}\alpha\cos\beta \pm \cos\alpha\operatorname{sen}\beta$

M.A.S

$\omega_0 = \sqrt{\frac{k}{m}}$ $\omega_0 = 2\pi f$ $T = \frac{2\pi}{\omega_0}$ $f = \frac{1}{T}$

$x(t) = A\cos(\omega_0 t + \varphi_0)$
$v(t) = -\omega_0 A sen(\omega_0 t + \varphi_0)$
$a(t) = -\omega_0{}^2 A\, cos(\omega_0 t + \varphi_0)$



$E = K + U = \frac{1}{2}KA^2$

Osc. Amortiguado $\gamma = \frac{b}{2m} = \frac{1}{2\tau}$

($\gamma > \omega_0$) Sobreamort. $x(t) = \left(Ae^{\sqrt{\gamma^2-\omega_0^2}\cdot t} + Be^{-\sqrt{\gamma^2-\omega_0^2}\cdot t}\right)e^{-\gamma t}$

($\gamma = \omega_0$) crit. Amort. $x(t) = Ae^{-\gamma t}(1+t)$

($\gamma < \omega_0$) Subamort. $x(t) = \underbrace{Ae^{-\gamma t}}_{A(t)}\cos(\omega' t + \varphi_0)$; $\omega' = \sqrt{\omega_0^2 - \gamma^2}$

Energia del subamort. $E = \underbrace{\frac{1}{2}KA^2}_{E_0}.e^{-2\gamma t}$
$\gamma << \omega 0$

Para péndulo: $\theta(t) = \theta_0\cos(\omega_0 t + \Phi_0)$
$\omega_0 = \sqrt{\frac{g}{L}}$ $S = L\theta$ $\ddot{S} = L\ddot{\theta}$ $v = \dot{\theta}L$

$\gamma T' = \frac{\pi}{\sqrt{Q^2 - \frac{1}{4}}}$ $\tau = \frac{1}{2\gamma}$ $Q = \frac{\omega_0}{2\gamma} = \omega_0\tau$ $T' = \frac{2\pi}{\omega'}$

- Crit. Q=1/2
- Sobreamort. Q<1/2
- Subamort. Q>1/2

Osc. Forzado:

$F(t) = F_0\cos(\omega_e t)$

$x(t) = A(\omega_e).\cos(\omega_e t - \delta)$

$$A(\omega) = \frac{F_0}{m\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + (2\gamma\omega)^2}} = \frac{F0}{\sqrt{(k - m\omega^2)^2 + (b\omega)^2}}$$

$$\tan\delta(\omega) = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \quad | \quad \operatorname{sen}\delta(\omega) = \frac{b\omega A(\omega)}{F_0} = \frac{2\omega\gamma A}{F0/m} = \frac{2\omega\gamma}{\sqrt{(\omega 0^2 - \omega^2)^2 + (2\omega\gamma)^2}}$$

$A_{max} \equiv A(\omega_{RA}) = \frac{F_0}{b\omega'}$ $\omega_{RA} = \sqrt{\omega_0^2 - 2\gamma^2}$

$\omega_{RP} = \omega_0$ $< P >= \frac{F_0^2}{2b}\operatorname{sen}^2(\delta(\omega)) = \frac{b\omega^2 A^2(\omega)}{2} = \frac{1}{2}F0.\omega A\, sen(\delta)$

$< P >_{max} \equiv < P > (\omega_{RP} = \omega_0) = \frac{1}{2}\frac{F_0^2}{b}$

$\Delta\omega = \omega 2 - \omega 1 \approx \frac{\omega 0}{Q} = \frac{1}{\tau} = 2\gamma$

Ondas propagantes no dispersivas:

$y(x,t) = A\cos(kx \pm kv + \varphi 0)$

$v = \sqrt{\frac{F}{\mu}}$ $\omega = kv$ $k = \frac{2\pi}{\lambda}$ $v = \lambda f = \frac{\omega}{k}$

$\lambda = v.T$ $T = \frac{2\pi}{\omega}$

$v\ max = A\omega$ $|a_{MAX}| = A\omega^2$

$P\ max = \mu\Delta x\, A\omega$ $F\ max = \mu\Delta x\, A\omega^2$

$< P >= \frac{1}{2}\mu\omega^2 A^2 v$

$\mathcal{U}_E = 2\mathcal{U}_K = 2\mathcal{U}_U = \mu A^2\omega^2\operatorname{sen}^2(kx \pm \omega t + \phi_0)$
$\mathcal{U}_E = \mathcal{U}_K + \mathcal{U}_U = \frac{1}{2}\mu\omega^2 A^2\left[\operatorname{sen}^2(kx)\operatorname{sen}^2(\omega t + \phi_t) + \cos^2(kx)\cos^2(\omega t + \phi_t)\right]$

Ondas estacionarias:

y(x,t)=A cos(kx+ϕ_x)cos(ωt+ϕ_t)

cuerdas con ext. Fijos: $L = n.\frac{\lambda_n}{2}$ $f_n = n.\frac{v}{2L}$ $(n\ entero)$ y(x,t)=A sen(kx)cos(ωt+ϕ_t)

cuerdas con ext. Libres: $L = n.\frac{\lambda_n}{2}$ $f_n = n.\frac{v}{2L}$ $(n\ entero)$ y(x,t)=A cos(kx)cos(ωt+ϕ_t)

cuerdas con ext. Mixtos: $L = n.\frac{\lambda_n}{4}$ $f_n = n.\frac{v}{4L}$ n impar
- si n arranca en un nodo: y(x,t)=A sen(kx)cos(ωt+ϕ_t)
- si n arranca en un antinodo: y(x,t)=A cos(kx)cos(ωt+ϕ_t)

$$u_k = \frac{1}{2}\mu\omega^2 A^2 sen^2(kx) sen^2(\omega t + \varphi_t)$$

$$u_u = \frac{1}{2}Fk^2 A^2 cos^2(kx) cos^2(\omega t + \varphi_t)$$

$$\mu\omega^2 = Fk^2$$

$E_{[x_i,x_f]} = \int_{x_i}^{x_f} u_E \, dx$ $E = \frac{1}{2}\mu\omega^2 A^2 \left(\frac{x_f - x_i}{2}\right)$ (si x_i;x_f son N o AN) $E_{\Delta x} \approx u_E.\Delta x$ (si $\Delta x \ll \lambda$)

Óptica:

$n \operatorname{sen}\theta = n' \operatorname{sen}\theta'$

$v = \frac{c}{n}$

Dioptra esférica: $\frac{n'}{s'} - \frac{n}{s} = \frac{(n'-n)}{R}$ s'>0 real , s'<0 virtual $A = \frac{h'}{h} = \frac{ns'}{n's}$

Lentes delgadas: $\frac{n''}{s'} - \frac{n}{s} = \frac{(n'-n)}{R_1} + \frac{(n''-n')}{R_2} = \frac{n''}{f'} = -\frac{n}{f}$ $A = \frac{h'}{h} = \frac{ns'}{n''s}$

f'>0 CV , f'<0 DV Potencia: $P = \frac{n''}{f'} = \frac{n''}{s'} - \frac{n}{s}$ (dioptrías si f' está en metros)

Lente inmersa en único medio: $\frac{1}{s'} - \frac{1}{s} = \left(\frac{n'}{n} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{1}{f'} = -\frac{1}{f}$ $A = \frac{h'}{h} = \frac{s'}{s}$ (si n"=n)

Espejos: $\frac{1}{s'} + \frac{1}{s} = \frac{2}{R}$ $A = \frac{h'}{h} = -\frac{s'}{s}$ s'>0 virtual , s'<0 real

Cristal	n_o	n_e
Calcita $CaCO_3$	1,6584	1,4864
Hielo H_2O	1,3090	1,3130
Cuarzo SiO_2	1,5443	1,5534
Zircón $ZrSiO_4$	1,9230	1,9680
Rutilo TiO_2	2,6160	2,9030

$A\cos(\omega t + \delta_1) + B\cos(\omega t + \delta_2) = C\cos(\omega t + \gamma)$ $C = \sqrt{A^2 + B^2 + 2AB\cos(\delta_1 - \delta_2)}$ $\tan\gamma = \frac{A\operatorname{sen}\delta_1 + B\operatorname{sen}\delta_2}{A\cos\delta_1 + B\cos\delta_2}$

$A\operatorname{sen}(\omega t + \delta_1) + B\cos(\omega t + \delta_2) = C\cos(\omega t + \gamma)$ $C = \sqrt{A^2 + B^2 + 2AB\operatorname{sen}(\delta_1 - \delta_2)}$ $\tan\gamma = -\frac{A\cos\delta_1 - B\operatorname{sen}\delta_2}{A\operatorname{sen}\delta_1 + B\cos\delta_2}$

$A\operatorname{sen}(\omega t + \delta_1) + B\operatorname{sen}(\omega t + \delta_2) = C\cos(\omega t + \gamma)$ $C = \sqrt{A^2 + B^2 + 2AB\cos(\delta_1 - \delta_2)}$ $\tan\gamma = \frac{A\operatorname{sen}\delta_1 + B\operatorname{sen}\delta_2}{A\cos\delta_1 + B\cos\delta_2}$

$$\int \operatorname{sen}^2(ax)dx = \frac{x}{2} - \frac{\operatorname{sen}(2ax)}{4a} \qquad \int \cos^2(ax)dx = \frac{x}{2} + \frac{\operatorname{sen}(2ax)}{4a}$$

$< f(t) >_T = \frac{1}{T}\int_t^{t+T} f(t')\,dt'$ $< \operatorname{sen}^2 x > = < \cos^2 x > = \frac{1}{2}$ $< \operatorname{sen} x > = < \cos x > = 0$